

DATE-04/12/14

Induction m/c

* These are the most commonly used motors because of their basic ad. like simple design, less cost, mech. strong, excellent running c/s, good speed regn no arm. reactions, commutation, sparkings & operates on AC (No need of any dc supply).

* RMF → It also operates as gen^r but not preferred conventional pwr generation these are more popular in motor segment.

* They operate on the basis of Rotating magnetic field.

* RMF → * When a 3 ϕ supply is applied to a 3 ϕ wdg which is distributed with the space displacement 120° a net flux is produced which rotates at a constant speed ($N_s = 120f/p$) with a const. magnitude ($1.5\phi_m$) and with a particular dirn depending on ϕ seq. -

$$\phi_A = \phi_m \sin \omega t$$

$$\phi_B = \phi_m \sin(\omega t - 120^\circ)$$

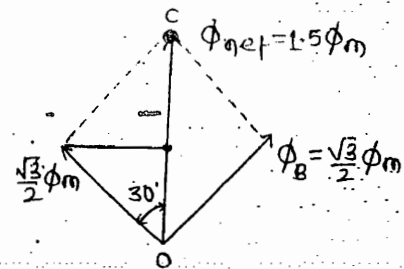
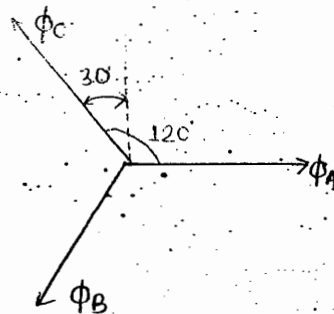
$$\phi_C = \phi_m \sin(\omega t - 240^\circ)$$

at $\omega t = 0$

$$\phi_A = 0$$

$$\phi_B = -\frac{\sqrt{3}}{2} \phi_m$$

$$\phi_C = \frac{\sqrt{3}}{2} \phi_m$$



From ΔOAB

$$\cos 30^\circ = \frac{OA}{OB}$$

$$OA = OB \cos 30^\circ = \frac{\sqrt{3}}{2} \phi_m \cdot \frac{\sqrt{3}}{2} = \frac{3}{4} \phi_m$$

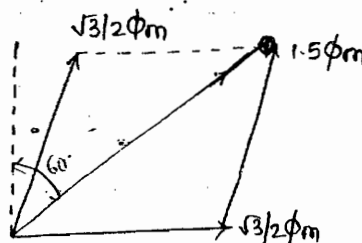
$$OC = \frac{3}{2} \phi_m = 1.5 \phi_m$$

at $\omega t = 60^\circ$

$$\phi_A = \frac{\sqrt{3}}{2} \phi_m$$

$$\phi_B = -\frac{\sqrt{3}}{2} \phi_m$$

$$\phi_C = 0$$

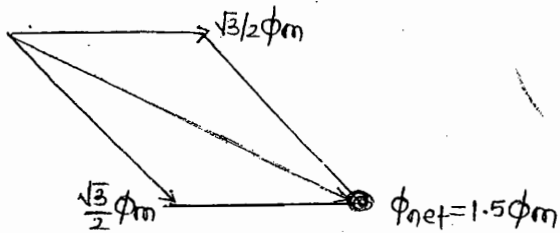


at $\omega t = 120^\circ$

$$\phi_A = \sqrt{3}/2 \phi_m$$

$$\phi_B = 0$$

$$\phi_C = -\sqrt{3}/2 \phi_m$$

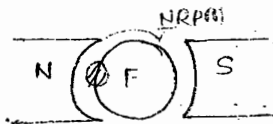
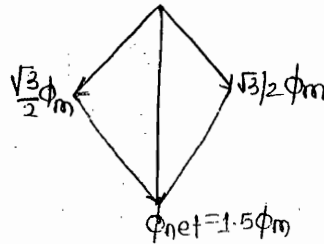


at $\omega t = 180^\circ$

$$\phi_A = 0$$

$$\phi_B = \sqrt{3}/2 \phi_m$$

$$\phi_C = -\sqrt{3}/2 \phi_m$$



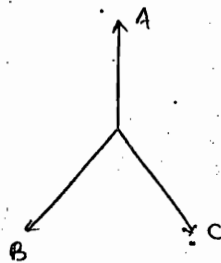
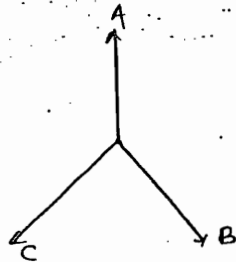
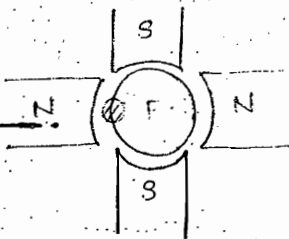
$N/60$ RPS

Cycles/rotation = $P/2$

$$\omega t / \text{sec} \times \text{cycles/rotation} = \frac{PN}{120}$$

$$\text{Cycles/sec} \Rightarrow f = \frac{PN}{120}$$

$$N_s = \frac{120f}{P}$$



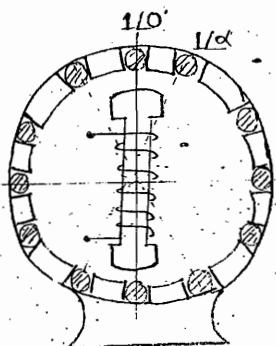
* Construction →

- * Like all rotating ele. m/c it also consists of stationary part stator & the rotating part rotor mounted on the shaft & supported by suitable bearings.
- * There will be least possible air gap b/n stator & rotor.

* Stator →

- * It contains slots at the inner peripheral made up of SI-steel laminations.
- * 3φ wdg either λ (or) Δ (depending on starting method) will be placed in the slots which offer good mech. security for the wdg.

* Design of 3φ wdg →



slots angle α°

Phase displacement contributed by 1 slot in ele°

$$\alpha = \frac{180^\circ}{\text{No. of slots/pole}}$$

$$\text{Slots} = 12, \alpha = 30^\circ$$

$$\text{Slots} = 120, \alpha = 3^\circ$$

Phase Group / Band / Belt : n (Adjacent)

Phase spread : $n\alpha$

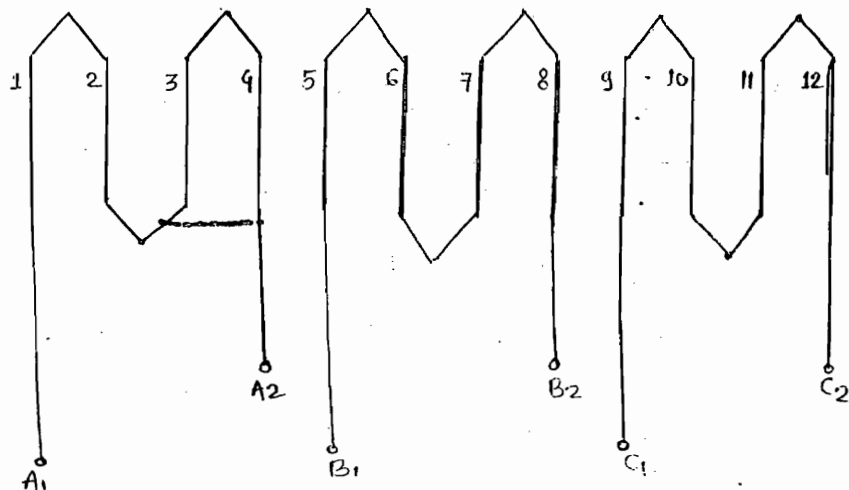
Phase group → The no. of adjacent slots (or) cond^r belonging to a same phase.

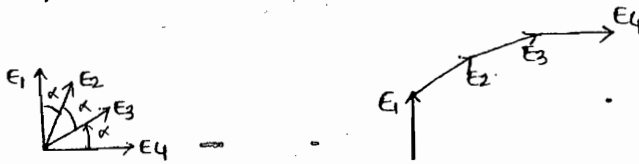
Phase spread → The angle subtended by one phase group is phase spread.

3φ wdg can be designed for 120° ph. spread (or) 60° phase spread.

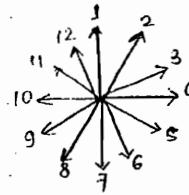
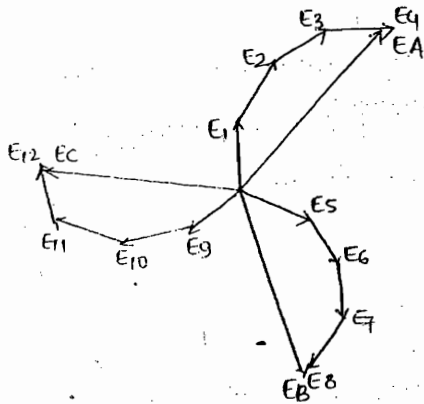
120 phase spread → $n\alpha = 120^\circ, \alpha = 30^\circ, n = 4$

$n \rightarrow$ slots/pole/phase (SPP)

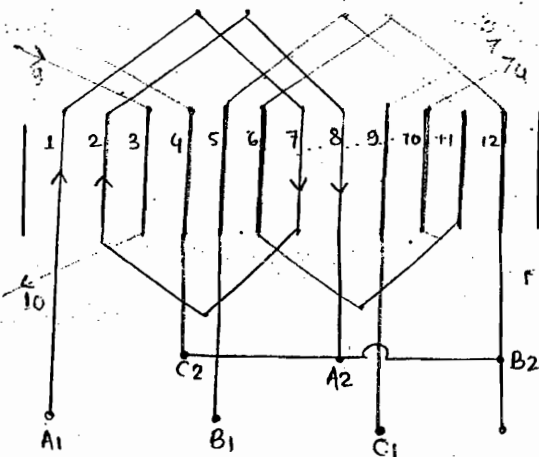




* Distribution of wdg is done only for getting improving the shape of waveforms.



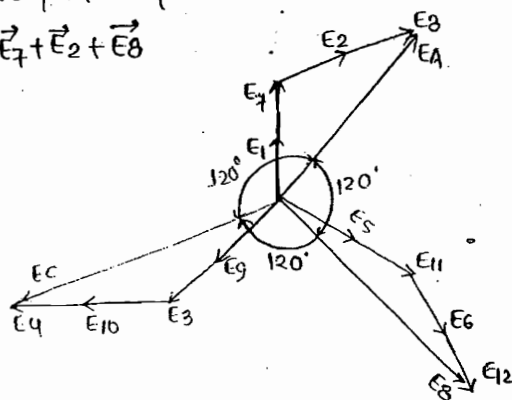
* 60° phase spread $\rightarrow n\alpha = 60^\circ; \alpha = 30^\circ; n = 2 \rightarrow 6$ phases
 $\downarrow$
 Combined to be 3 phase
 Bunch of 2 phases.



Notes:-

To maintain 60° phase spread 1 should be connected to 7.

$$E_A = \vec{E}_1 + \vec{E}_7 + \vec{E}_2 + \vec{E}_8$$



$$E_{120^\circ} < E_{60^\circ}$$

Phase spreads

distribution factor $k_d = \frac{\text{Emf } Dw}{\text{Emf } cw}$

$$k_d = \frac{\sin \frac{n\alpha}{2}}{n \sin \frac{\alpha}{2}}$$

since; $\sin \frac{\alpha}{2} \approx \frac{\alpha}{2}$

$$k_d = \frac{\sin \frac{n\alpha}{2}}{\frac{n\alpha}{2}}$$

$$k_{d60} = \frac{\sin \frac{60}{2}}{60/2}; \quad k_{d120} = \frac{\sin \frac{120}{2}}{120/2}$$

$$\frac{k_{d60}}{k_{d120}} = 1.15$$

$$k_{d60} = 1.15 k_{d120}$$

* Pitch Factor $\rightarrow$

$$k_p = \cos \beta/2$$

$$3^{rd} \rightarrow k_p = 0.866$$

$$5^{th} \rightarrow k_p = 0.95$$

$$7^{th} \rightarrow k_p = 0.97$$

$$k_{p3} = \cos 3\beta/2$$

$$k_{p7} = \cos 7\beta/2$$

$$\frac{7\beta}{2} = 90^\circ$$

$$\beta = \frac{180}{7}$$

$$k_{d7} = \frac{\sin \frac{n\alpha}{2}}{n \sin \frac{\alpha}{2}}$$

$$\sin \frac{n\alpha}{2} = 180^\circ$$

$$n\alpha = \frac{360^\circ}{7}$$

* Practically 60° phase spread is preferred as it gives better vol. & power ratings. 120° pha. spread will eliminate 3rd harmonic content in line.

However the wdg are preferred to be λ . Therefore the 3rd harmonic content in line is 0 even though it exist in each phase.

* Harmonics can be eliminated by short pitching the coils by introducing Pitch factor eliminating higher order harmonics like 5th & 7th is more advantageous.

* For high vol. λ connected stator wdg is preferred as it reduces insulation requirements as well as no. of turns in each phase.

* The 3 ϕ wdg is also classified acc to 2 types :-

(1) Integral slot wdg (2) Fractional slot wdg

$n = \text{SPP (Integer)}$

Eg:- $12/2/3$, $18/2/3$

$n = \text{non integer (Fraction)}$

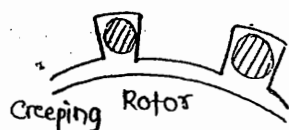
Eg:- $18/4/3$

* Fractional slot wdg practically offer short pitched wdg specially more suitable in double layer wdg cases.

Fractionally $\rightarrow$ Automatically short pitched

* Type of slots $\rightarrow$

Open

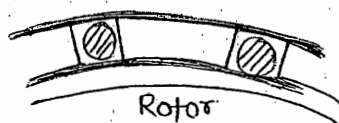


* Wdg design, routine maintenance, repairs is easy
less expensive.

* Offer high reluctance to leakage flux, less leakage reactance.

* Net air gap is more which requires more magnetising current to maintain flux & the operating PF will be low wrt λ .

Closed

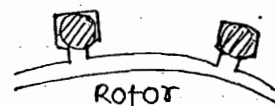


Difficult

High leakage flux, high leakage reactance

Net air gap is less, so magnetising current less
PF more.

Semi-open/closed



quite difficult

moderate.

moderate.

* Wrt leakage reactance its
Operating pf is better
Comparitively.

Wrt leakage reactance
Operating pf is low

moderate.

* Non uniform air gap will
produce space harmonics
by distorting flux wave-
form. also called as slot/
tooth harmonics.

No slot/tooth harmonics

moderate

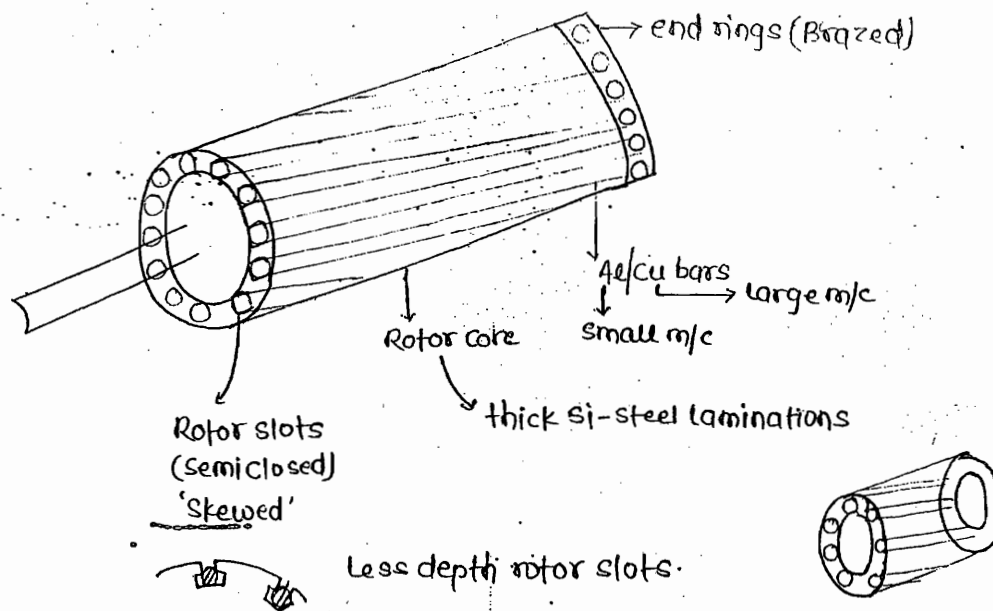
* Generally preferred slots in IM in stator (OR) rotor are semiclosed type.

* Large rating m/c eg power plant gen (OR) large rating IM preferred open type
because they produce less leakage reactance & also offers easy, maintaince
repair works.

* Rotor →

(1) Squirrel cage (OR) cage rotors

(2) Wound / Slip Ring rotors.



* It has most simple construction doesn't contain any wdg.

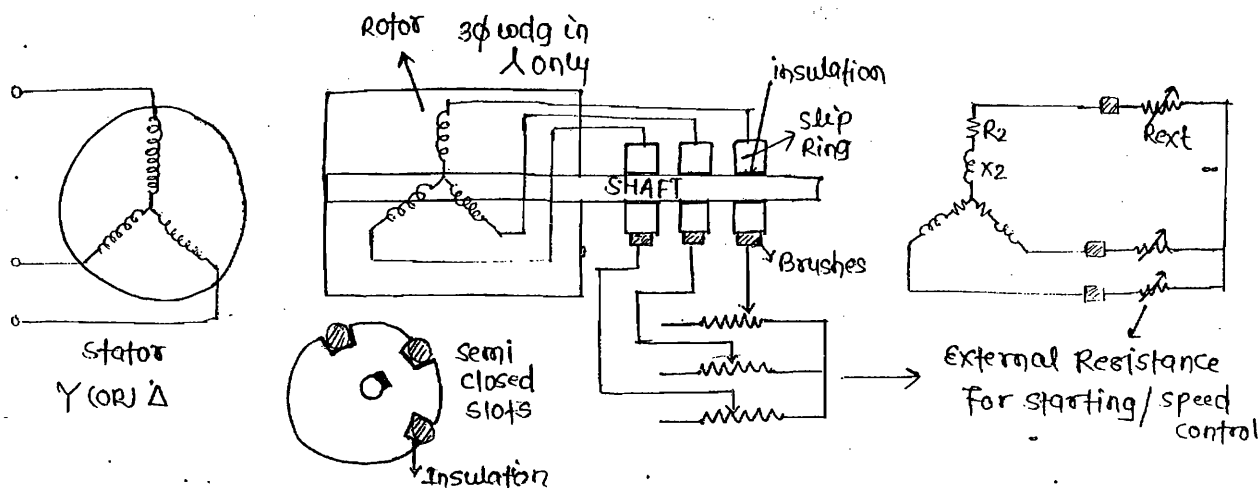
* Rotor is made up of thick Si-steel laminations punched into slots of less depth
Preferably semiclosed.

* Al (OR) Cu bars are directly placed in this slots w/o any insulation (as the current
flows only through least resistance)

- * These bars are solidally closed (or) sc with 2 end rings (Brazed)
- * Rotor slots are skewed (twisted or inclined) wrt shaft axis or stator slots to avoid locking tendency during starting.
- * ^{Under} the running condⁿ rotor freq. is negligible & the rotor core losses are very low.
- Therefore no need of thin laminations.
- * It requires least maintenance repairs, mech. strong, simple design, less cost & excellent running c/s.
- * The drawback is due to its low starting torque.
- * As the rotor is permanently closed its resistance can't be varied for starting (or) controlling purpose.
- “Poles are automatically formed on the rotor equal to the no. of poles on stator.”
- * Due to mutual indⁿ the poles are induced in the rotor.
- Therefore it reacts to any no. of poles on the stator.

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- (2) Slip Ring/wound → * The rotor is a cylindrical drum shaped st. punched into slots (semiclosed) on the outer peripheral which contain a 3 ϕ wdg similar to stator wdg but essentially star connected.
- * Rotor is designed for the same no. of poles as on the stator. but preferably the no. of slots should not be same & should not have a common factor to avoid any chance of magnetic locking during starting.
 - * The 3 terminals of Δ connected wdg will be brought out & connected to 3 slip rings mounted on the same shaft with suitable insulation.



* The starting torque is $\propto$ to rotor resistance but the rotors are naturally designed for least possible resistance. In order to have high starting torque an external resistance is inserted in each phase equally through slip ring & brushes.

* Under running condⁿ the resistance is disconnected using a METAL COLLAR arrangement the brushes are slight down & 3 slip rings are sc to form a closed rotor.

* This is only used for high starting torque appⁿ. Around 90% of motors are squirrel cage type only.

* Slip motors are expensive, complicated design, high maintainance repairs, running c/s are not good comparatively.

* Principle:- (How rotor run)?

* When a 3 ϕ supply is given to stator RMF is produced at N_s . As the flux sweeps past the rotor cuts it & induces EMF due to relative speed N_s . & the rotor freq. at stand still is supply freq.

* As the rotor is essentially closed it produces current & the current carrying cond^r placed in the magnetic field will experience a force which is torque & the rotor rotates.

* According to Lenz law it rotates in the dirⁿ of magnetic field to oppose the cause i.e. relative speed.

* Actually the rotor want to catch the rotating magnetic field but it could not catch it due to losses in the rotor. and rotates at a speed N slightly less than N_s .

* The rotor slips back RMF by a speed $N_s - N$ known as slip speed.

* The principle of operation is mutual indⁿ acc. to Faraday & it is equivalent to a rotating Xmer with sc 2 o .

* It is a singly excited m/c.

* Vol. is applied only to stator & the vol. are induced in the rotor through electromagnetic indⁿ.

Therefore called as indⁿ motor.

* As it can't run at synchronous speed it is also called as asynchronous motor.

* slip $\rightarrow$

* It is the diff of syn. speed & actual speed of the rotor expressed in % of syn. speed.

$$S = \frac{N_s - N}{N_s}$$

$$\%S = \frac{N_s - N}{N_s} \times 100$$

$$N_s - N = SN_s$$

$$N = N_s(1 - S)$$

* During starting at $N=0$, $S=1.0$

* A well designed indⁿ motor runs near to syn. speed with low value of slip on NL & slip is near to 0 (but not 0)

* If the rotor runs at N_s , then no relative speed, no E_2 , $I_2=0$, $T=0$.

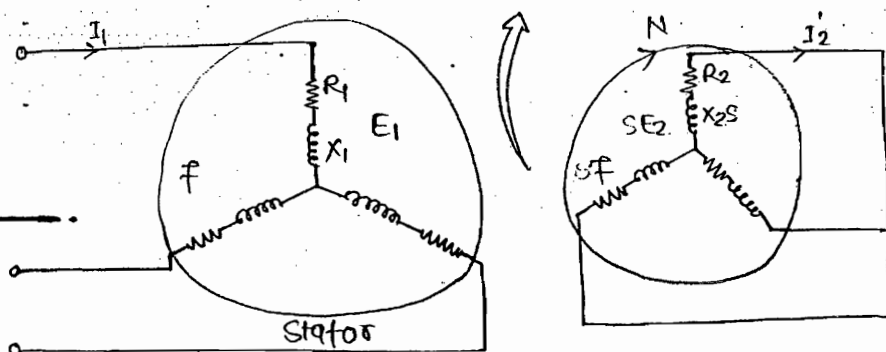
* The operating region of indⁿ motors will be around 1-5% slips from NL to rated load.

* For small motor it may be around 8% at rated load.

* slip plays major role in the operation & performance & behaviour of Im.

* Affect of slip $\rightarrow$

Representation $\rightarrow$



$$E_2 \propto N_s - 0$$

$$E_2' \propto N_s - N$$

$$\frac{E_2'}{E_2} = \frac{N_s - N}{N_s}$$

$$E_2' = SE_2$$

$$\text{Relative Speed} = \frac{N_s - N}{N_s}$$

* If the rotor resistance is adjusted to a suitable value i.e. exactly equal to rotor leakage reactance at standstill then the motor starts with its max^m torque.

* Under such condⁿ the rotor PF will be 0.707 lagging

* Running torque → Consider a motor running at rated load with slip s & speed N_s

$$T_f \propto E_2' I_2' \cos \phi_2'$$

$$T_f \propto [K_1 \phi] I_2' \cos \phi_2' \quad (E_2 \propto \phi)$$

$$T_f \propto (K_1 \phi) \times \frac{SE_2}{\sqrt{R_2^2 + (SX_2)^2}} \cdot \frac{R_2}{\sqrt{R_2^2 + (SX_2)^2}}$$

$$T \propto \frac{(K_1 \phi) SE_2 R_2}{[R_2^2 + (SX_2)^2]}$$

$$T = K \cdot \frac{SE_2^2 R_2}{R_2^2 + (SX_2)^2}$$

$$\left(K = \frac{3 \times 60}{2\pi N_s} \right)$$

* For a given (or) const. supply vol^t: $T_f \propto \frac{SR_2}{R_2^2 + (SX_2)^2}$

* Under normal running condⁿ at low values of slip;

$$s \downarrow; (SX_2)^2 \ll R_2$$

$$T_f \propto \frac{SR_2}{R_2^2}$$

$$*** \boxed{T_f \propto \frac{s}{R_2}}$$

& at the high values of slip

$$s \uparrow (SX_2)^2 \gg R_2$$

$$T_f \propto \frac{SR_2}{(SX_2)^2} \propto \frac{R_2}{SX_2^2}$$

$$\boxed{T_f \propto \frac{R_2}{SX_2^2}}$$

* For a given R_2, X_2 values of rotor the torque developed is directly proportional to slip in the low slip region & torque is inversely proportional to slip in the high slip region.

* The running torque are also sensitive to supply vol. variations.

* Condⁿ for max^m running torque →

$$T_F = \frac{SR_2}{R_2^2 + (SX_2)^2}$$

$$\text{let } y = \frac{1}{T_F} \uparrow$$

$$\frac{dy}{ds} = 0$$

$$y = \frac{1}{T_F} = \frac{R_2^2 + (SX_2)^2}{SR_2}$$

$$\frac{dy}{ds} = -\frac{R_2}{s^2} + \frac{X_2^2}{R_2} = 0$$

$$\frac{R_2}{s^2} = \frac{X_2^2}{R_2}$$

$$R_2^2 = X_2^2 s^2$$

$$\boxed{R_2 = SX_2}$$

* If the rotor resistance = slip times the rotor reactance at stand still then the rotor develops max^m running torque.

$$\text{max}^m \text{ starting torque } \boxed{R_2 = X_2}$$

$$\text{max}^m \text{ running torque } \boxed{R_2 = SX_2}$$

$$\begin{array}{c} \text{Starting} \\ T_{st\max} \propto \frac{R_2}{R_2^2 + X_2^2} \end{array}$$

$$T_{st\max} \propto \frac{1}{2X_2}$$

$$\begin{array}{c} \text{Running torque} \\ T_{F(\text{running})} \propto \frac{SR_2}{R_2^2 + (SX_2)^2} \end{array}$$

$$T_{F(\max)} \propto \frac{1}{2X_2}$$

$$\boxed{T_{\max} \propto \frac{1}{2X_2}}$$

$$K_1 = \frac{3 \times 60 E_2^2}{2\pi N_s}$$

(Refer $\frac{V}{f}$ speed control)

$$\boxed{s_m = \frac{R_2}{X_2}}$$

(Also a condⁿ for max^m value)

* The motor rotates at any slip b/n 0 & 1 but it delivers its max^m torque only at a particular slip $s_m = R_2/X_2$.

* Generally rotor resistance is a least possible value compared to its leakage reactance at stand still.

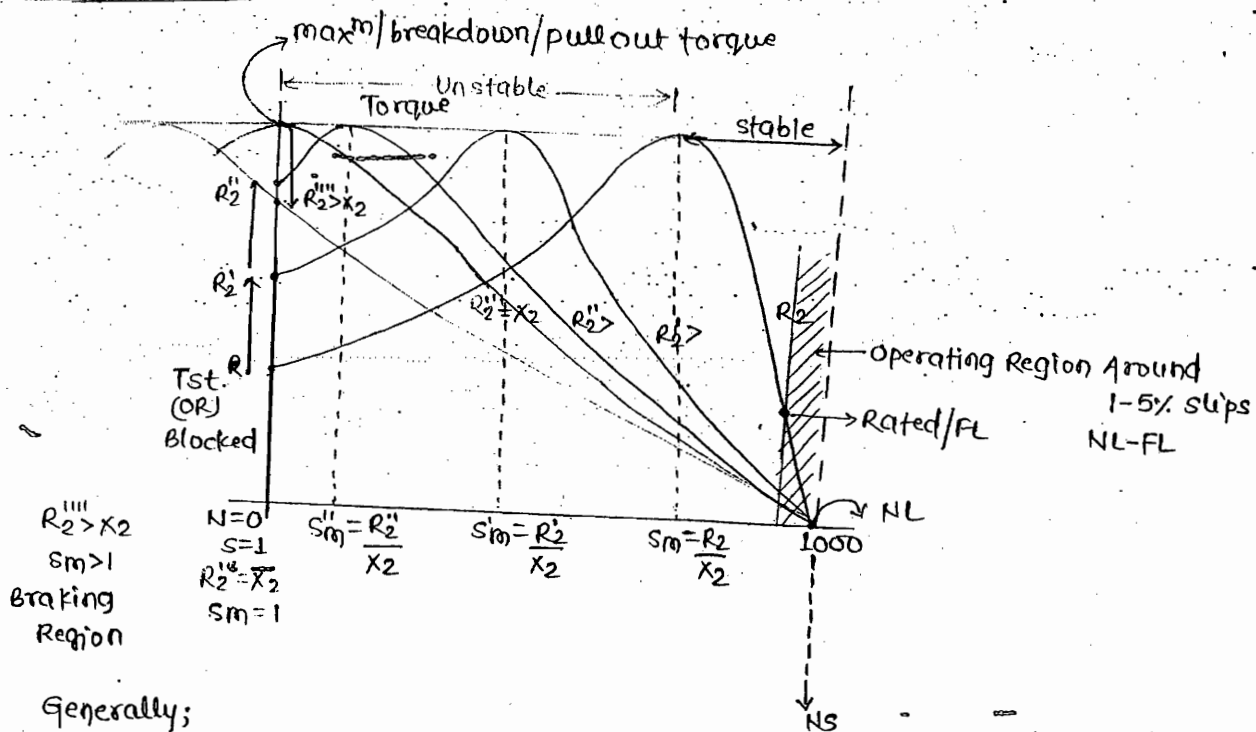
* If an external resistance is added into the rotor to make $R_2 + R_{ext} = X_2$ then $s_m = 1$ which means the motor starts with its max^m torque because slip at which max^m torque occurs $s_m = 1$.

* max^m torque magnitude is independent of rotor resistance but the slip at which it occurs depends on R_2 .

* max^m torque is inversely proportional to leakage reactance of rotor at stand still.

Therefore it will be designed as min^m as possible with less depth slots.

* Torque-slip c/s →



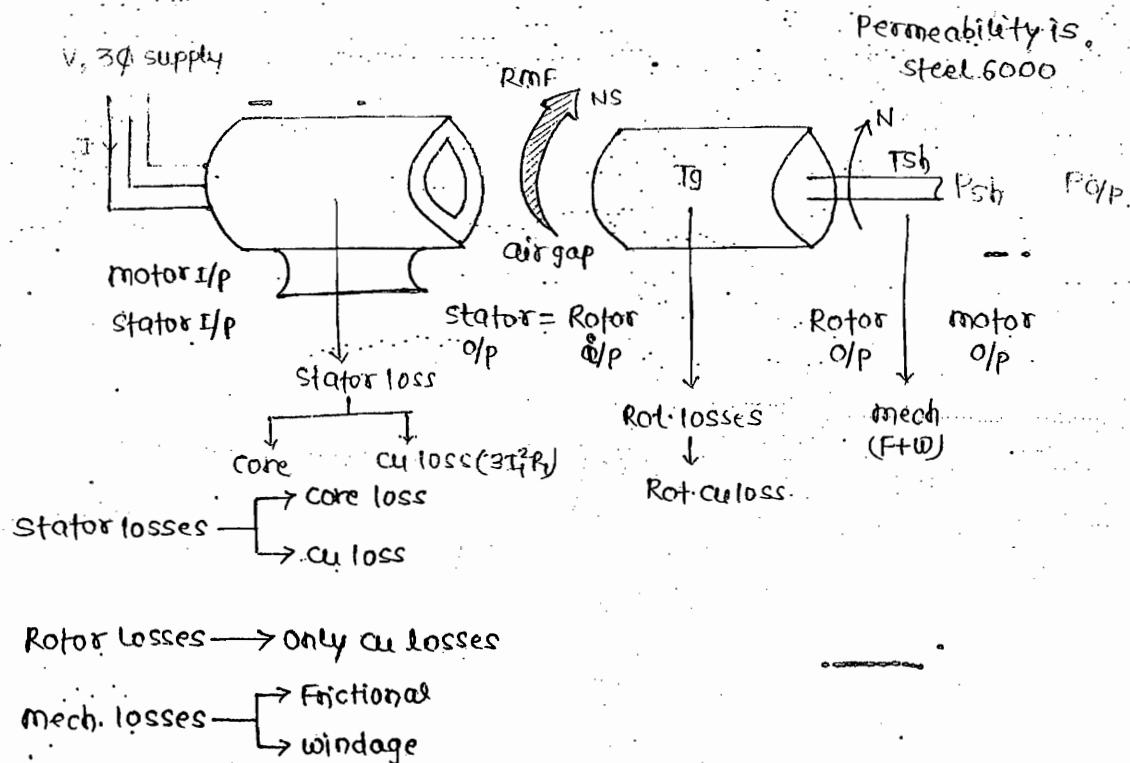
• $T_{max} \gg T_f$

($T_{max} \approx 2-3 T_f$)

• ($T_{st} \approx 1.5 T_f$)

- * T-s c/s are essentially straight line in the operating region (Low slip). Every motor will have its $\max^m$ capacity (or) capability to develop torque which occurs only at slip s_m if the load increases beyond it it can't deliver torque & it stops which is unstable region.
- * Im have excellent overload capability but not desirable to operate at overloading but can be done for short duration if req.
- * All the stable region is not operating region, it should be operated at low slip only due to η consideration.
- * The $\max^m$ torque is independent of rotor resistance but the slip & speed where it occurs depends on rotor resistance.
- * If $R_2 \gg X_2$ $\max^m$ torque occurs in the braking region.

* Power stages $\rightarrow$



* There are no losses in the air gap.

$$\text{Stator I/p} - \text{stator loss} = \text{stator o/p} = \boxed{\text{Rotor I/p} = \text{Air gap power}}$$

$$\text{Rotor I/p} - \text{Rotor cu loss} = \text{Rot. o/p}$$

$$\text{Rot. o/p} - \text{Rot loss} = \text{shaft (or) motor o/p}$$

$$\eta = \frac{\text{motor o/p (Psh)}}{\text{motor I/p} (\sqrt{3} V_L I_L \cos \phi)}$$

$$P = \frac{2\pi N T_g}{60}$$

$$\text{Rotor P}_{i/p} = \frac{2\pi N_s T_g}{60} \text{ watts} \quad \left| \quad \text{Rot P}_{o/p} = \frac{2\pi N T_g}{60} \text{ watts} \right.$$

$$\text{Rotor P}_{i/p} - \text{Rotor P}_{o/p} = \text{Rot cu loss} = \frac{2\pi T_g}{60} (N_s - N) = \frac{2\pi T_g}{60} \cdot s = \frac{2\pi T_g N_s}{60} \cdot s$$

$\underbrace{\frac{2\pi T_g N_s}{60}}_{R_{i/p} \times s}$

$$\text{Rotor cu loss} = s \cdot \text{Rot. P}_{i/p}$$

$$\text{Rotor P}_{i/p} - \text{Rotor P}_{o/p} = s \text{ Rot. I/p} \Rightarrow \text{Rotor} = (1-s) \text{ Rotor P}_{i/p}$$

$$\frac{\text{Rotor P}_{o/p}}{\text{Rotor P}_{i/p}} = (1-s) = \eta_{\text{rotor}} = \frac{N}{N_s} \quad (\text{Approx } \eta \text{ of IM.})$$

(Negl. mech. & stat)

$$* \text{Core loss (iron)} + \text{mech (F + w)} = \text{Constant loss.}$$

* If IM is running at 950 rpm then its approx. η will be 90% because N_s taken as 1000 rpm in Indian condⁿ like (F $\rightarrow$ 50Hz etc). (Neglecting const. losses)

* T developed $\rightarrow$

$$T_g = \frac{60}{2\pi N_s} \times \begin{matrix} \text{Rotor power I/p} \\ \text{(OR)} \\ \text{Rotor cu loss} \end{matrix}$$

Rot. cu loss = Air gap power.

$$T_g = \frac{60}{2\pi N_s} \cdot \frac{\text{Rot. cu loss}}{s} \longrightarrow 3 I_2'^2 R_2$$

$$= \frac{60}{2\pi N_s} \cdot \frac{3 I_2'^2 R_2}{s}$$

$$= \frac{60}{2\pi N_s} \cdot 3 \times \left(\frac{s E_2}{\sqrt{R_2^2 + (s X_2)^2}} \right)^2 \times \frac{R_2}{s}$$

$$\therefore I_2' = \frac{s E_2}{\sqrt{R_2^2 + (s X_2)^2}}$$

$$= \frac{60}{2\pi N_s} \times \frac{3 \cdot s E_2^2}{R_2^2 + (s X_2)^2} \cdot R_2$$